

ASSESSMENT TOOL-3

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CO2 - ASSESSMENT TOOL - 3

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1. Footprinting Tools - Conducting Competitive Intelligence.

A passive security assessment involves collecting publicly available information about a target to organization without directly interacting with its systems.

1. Domain Registration Details

Tool : WHOIS

⇒ Retrieves domain registration information.

⇒ Displays:

- * Domain owner. (if public)
- * Register name
- * Registration and expiry dates
- * Name servers
- * Administrative and technical contact the details.

2. Employee Email Addresses - the Harvester

⇒ collects publicly available email addresses from search engines and public sources.

⇒ Find emails patterns used by employees.

⇒ use : Helps understand the technological used by the organization.

Identify employee contacts and understand the organization's structure.

3. Technology Stack - Wappalizer

- ⇒ Detects technologies used on the company's website.
- ⇒ Identifies web servers, CMS, frameworks, programming languages, and analytics tools.
- ⇒ Use: Helps understand the technologies used by the organization.

4. Publicly Exposed Subdomains - sublist3r

- ⇒ Discovers public subdomains using search engines and certificate records.
- ⇒ Examples: mail.company.com, portal.company.com, api.company.com.
- ⇒ Use: Helps identify publicly accessible services and the external attack surface.

Conclusion:

WHOIS, theHarvester, Wappalizer, and sublist3r help gather public information safely and support competitive intelligence and security assessments.

2. DNS Zone Transfer

A DNS zone transfer (AXFR) copies DNS records from a primary DNS server to a secondary DNS server. If the server is misconfigured, anyone may retrieve the DNS records.

Command :

Using dig :

```
dig AXFR example.com @ns1.example.com
```

or using nslookup :

```
nslookup
```

```
server ns1.example.com
```

```
ls -d example.com
```

If the zone transfer is successful

The following records may be obtained :

- ⇒ A records - IP addresses of hosts
- ⇒ MX records - Mail server information.
- ⇒ NS records - Name servers.
- ⇒ CNAME records - Alias records.
- ⇒ TXT records - Additional DNS information.

Use During Reconnaissance

- ⇒ Identifies internal hosts and servers.
- ⇒ Finds mail servers used by the organization.
- ⇒ Helps map the organization's network structure.

Conclusion:

DNS zone transfers should only be allowed for authorized DNS servers. Restricting AXFR prevents attackers from collecting sensitive network information.

3. Social Engineering - Social Engineering Attacks & Countermeasures

Type of Attack:

The attack is phishing attack. The attacker sends a fake email pretending to be the IT department and asks employees to verify their login credential through a fraudulent link.

Countermeasures:

1. Verify the sender's email address
Check whether the email comes from the official company domain.
2. Do not click suspicious links
However over the link to verify the URL before (responding). clicking.
3. Confirm with the IT department:
Contact the IT team through official channels before responding.
4. Never share login credentials
Do not enter usernames or passwords on unknown websites.
5. Report Suspicious Emails
Inform the IT or security team immediately.
6. Enable Multi-factor Authentication (MFA)
Use an additional verification method to protect accounts.

7. Follow security awareness practices.
Attend cybersecurity training and stay alert to phishing attempts.

How These Actions Help:

- ⇒ Prevent users from accessing fake websites.
- ⇒ Protect usernames and passwords from attackers.
- ⇒ Reduce the risk of unauthorized account access.
- ⇒ Improve the overall security of the organization.

Conclusion:

This scenario is an example of a phishing attack. by verifying emails, avoiding suspicious links, reporting phishing attempts, and using MFA, employees can prevent credential theft and protect organizational data.